Agentic Public Records Compliance Assistant: Final Capstone Report

Course: Carnegie Mellon University, Building Agentic AI Systems. Capstone, Research Assistant track. **Date:** November 2026. **Final evaluation commit:** `7a1a875`.

Evidence rule for this report. Every number, trace excerpt and screenshot comes from a run that was actually executed and saved in the repository. Where a design element was proposed but not built, or built differently, the report says so. Status labels used throughout:

Label	Meaning
Implemented	Code exists, runs, and has tests or recorded runs
Partially implemented	Code exists, but not all of it has been run or exercised
Changed	Built differently from the proposal
Proposed / deferred	Not built

1. Executive summary

The Agentic Public Records Compliance Assistant helps a public-records reviewer prepare a synthetic record for release under New Jersey's Open Public Records Act (OPRA). It runs as a sequence of steps:

1. **Detect.** The Record Analysis Agent finds candidate personal information.
2. **Retrieve.** The Policy Research Agent retrieves the governing text from five official nj.gov policy documents.
3. **Recommend and check.** The Compliance & QA Agent proposes redactions, validates every citation and classifies the evidence.
4. **Route.** The Orchestrator Agent routes each case to the standard ReAct path, a targeted retrieval, or a bounded Tree of Thought (ToT) subgraph.
5. **Human review.** The workflow stops for a person. A named reviewer approves, rejects, modifies or sends items back for re-evaluation. Only after explicit authorization is a deterministic redacted derivative written, with a hash-chained audit trail. Nothing is published.

What was built. The four agents run on a LangGraph `StateGraph`. All tools are served by an MCP server built with the official MCP Python SDK. ToT is a conditional subgraph, not a fifth agent.

The agents are deterministic and rule-based. No LLM is called anywhere in the workflow. The "ReAct" behaviour is a bounded, rule-driven decide → call tool → observe loop. A Claude planner was built in Milestone 3 but was tested only with a fake client.

Evaluation. 22 synthetic scenarios were run under two configurations, ReAct-only (A) and ReAct + selective ToT (B), for 44 runs:

Result	A	B
PII detection precision	1.000	1.000
PII detection recall	0.895 (17/19)	0.895 (17/19)
Unsupported redactions	0	0
Escalation recall	11/11	11/11
End-to-end success	22/22	22/22

- **What ToT changed:** exactly one decision. In S12 (a retired judge's home address) it selected a fully verified, cited redaction instead of abstaining. Cost: 13 extra tool calls and about 0.11 s per ToT case.
- **Before evaluation-driven fixes, ToT was harmful:** it made two unsupported redactions. Comparing A against B exposed four defects, which are now fixed and covered by regression tests.
- **Safety:** 16 of 16 safety checks passed over all runs, 40 of 40 mapped guardrail tests passed, and the full suite passed 314 of 314 tests.
- **Conclusion:** selective ToT did no harm after the fixes and helped one ambiguous case. The evidence does not support a general accuracy claim.

2. Problem statement and intended users

The task. Before a public record is released under OPRA, a custodian must:

- find personal information such as personal phone numbers, driver licence numbers, home addresses and IP addresses;
- determine whether the statute or Government Records Council (GRC) guidance requires redaction;
- redact consistently and record the basis.

The work is manual and repetitive. Errors cut both ways: a missed value is a privacy breach, and an unsupported redaction wrongly withholds public information.

Intended users. Public-records custodians and review staff, as stated in the original proposal.

The assistant's role is to reduce repetitive work while "keeping the final decision with a human reviewer" (proposal). It finds candidates, retrieves and cites the governing text, explains uncertainty, and prepares the derivative. It never decides disclosure.

Scope. Synthetic text records only. Public nj.gov policy documents only. No real records, personal data, evidence-management integrations or release operations.

3. Why a standalone LLM is insufficient

A single prompt to an LLM	What the task requires	How this system provides it
May produce a plausible but non-existent statute or section	Every redaction tied to authoritative text	Recommendations can cite only chunk ids returned by retrieval. `validate_citations` checks that each chunk exists, the excerpt is verbatim, and the source is official. The QA agent vetoes invalid citations
Knows only its training data	Current statute text (OPRA as amended by P.L. 2024, c.16)	Retrieval over five official nj.gov documents ingested with provenance
Cannot act	Detection, retrieval, validation, redaction	Tools exposed through MCP
One pass; no recovery	Retries, fallbacks, escalation	Bounded loops: ≤ 2 detection attempts, ≤ 2 queries per category, one targeted retry, step limit 30, tool timeouts
May follow instructions embedded in documents	Documents are evidence, not instructions	Rule-based agents never execute passage text; a provenance guard prevents non-official text from being cited
No record of why	Auditable decisions	Typed state, tool-call history, ToT node tables, hash-chained review events, audit report
Implies a decision	Human authority over disclosure	LangGraph `interrupt()` before completion; finalization requires a named reviewer and explicit authorization

4. Evolution from Checkpoint 1.1 through Checkpoint 5.1

Full mapping, with gaps: checkpoint_traceability.md.

Checkpoint	Decision	Outcome in the final system
1.1 Proposal (assumed to be the project proposal document)	Research assistant for OPRA redaction of video, audio and documents; policy-backed reasons; approve, reject or modify; escalate when unclear	Implemented for text; audio and video deferred
2.1 Reasoning, memory, tools (known only from 4.1's summary)	ReAct; short-term and long-term memory; external tools	Rule-based ReAct loop; typed state; ChromaDB + BM25 index
3.1 Executed retrieval	Real nj.gov ingestion; the phone-number query; agent over MCP; missing evidence	Reproducible (`python -m app.checkpoint_3_1`). 47:1A-5(a) retrieved at rank 2 (cosine 0.6537). Re-verified in Milestone 7
4.1 Tree of Thought	Two versions were written: "do not use ToT" and "use ToT selectively". The selective version was implemented: 3 / 3 / 2 beam search, 30/30/25/15 rubric, minimum 70	Implemented as a LangGraph subgraph; the critic is deterministic instead of a CrewAI/LangChain critic agent
5.1 Multi-agent architecture	Orchestrator plus three workers; hybrid graph;	Implemented on LangGraph; research

Checkpoint	Decision	Outcome in the final system
	structured shared state; QA can request more evidence	runs after detection (it depends on the findings) and in parallel across categories

The ToT reversal. The 4.1 drafts record a real design debate. The first draft argued that the problem is evidence, not breadth of reasoning, and that more branches could mean more unsupported answers. The final design adopts ToT *selectively*, guarded by mandatory evidence checks. The evaluation partly vindicated the first draft: before the fixes, ToT's extra branches produced two unsupported redactions that the ReAct path did not.

5. Overall logical architecture

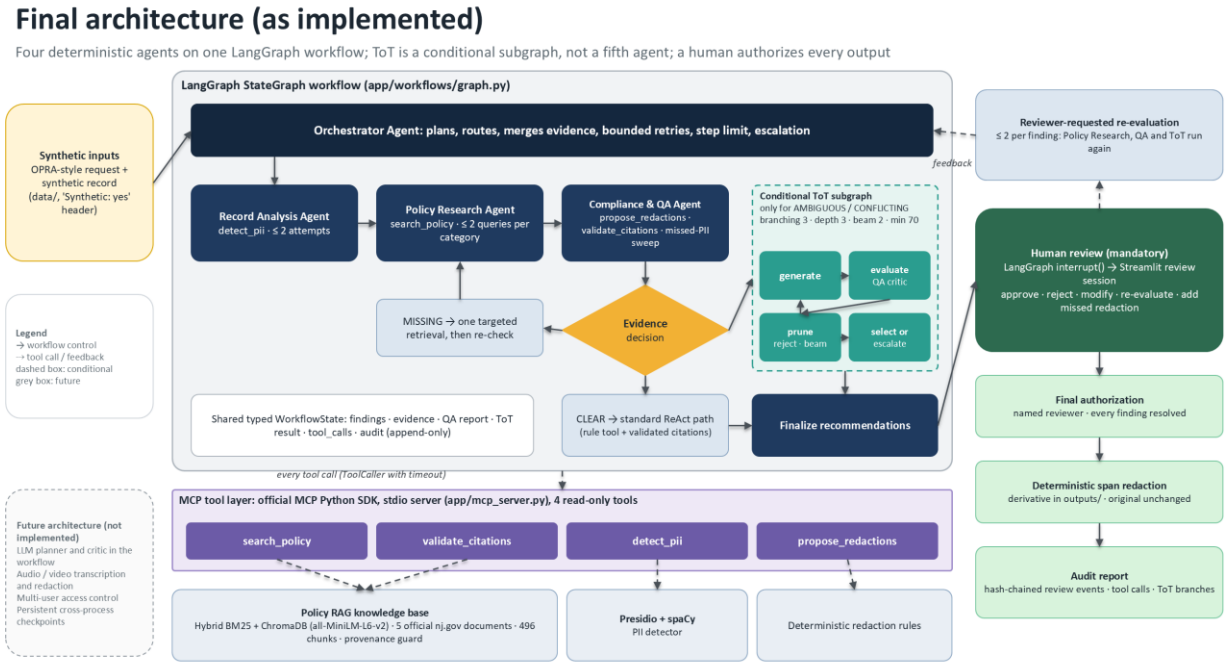


Figure 1. Final architecture as implemented. Editable sources: `architecture/final_architecture.drawio`, `architecture/final_architecture.pptx`, and Mermaid diagrams in `architecture/README.md`.

Layer	Components
Inputs	Synthetic OPRA-style request and record (`data/`). Every file carries a `Synthetic: yes` header, and the loader refuses files without it
Workflow	LangGraph `StateGraph` (`app/workflows/graph.py`) with a typed `WorkflowState`. Nodes: `orchestrator_start`, `record_analysis`, `policy_research`, `evidence_merge`, `compliance_qa`, `evidence_decision`, `targeted_retrieval`, `error_recovery`, `tot` (subgraph), `finalize_recommendations`, `human_review` (interrupt), `complete`
Agents	Orchestrator, Record Analysis, Policy Research, Compliance & QA (`app/workflows/agents.py`, `graph.py`)
Reasoning	Standard path for CLEAR evidence; targeted retrieval for MISSING; ToT subgraph (`app/workflows/tot.py`) for AMBIGUOUS or CONFLICTING
Tools	MCP server (`app/mcp_server.py`): `detect_pii`, `search_policy`, `propose_redactions`, `validate_citations`
Knowledge	Policy index (`app/policies/`): 5 documents, 496 chunks, BM25 + ChromaDB; Presidio + spaCy detector; deterministic redaction rules
Human review	`app/review/` (session, re-evaluation, redaction, report) and the Streamlit app `app/review_app.py`
Outputs	Derivative in `outputs/derivatives/` (never the original path); review sessions in `outputs/reviews/`; audit report JSON

6. MCP tool integration

Status: Implemented.

The server is app/mcp_server.py. It uses the official MCP Python SDK 2.3.0 (MCPServer) over stdio. Its four tools wrap one validated ToolService (app/tools/service.py). Configuration comes from environment variables: PRA_DATA_DIR, PRA_SEMANTIC, PRA_POLICY_DOC_IDS and PRA_LOG_LEVEL.

Tool	Input → output	Called by	Notes
detect_pii	record id → findings (category, span, text, confidence)	Record Analysis	Presidio 2.2.364 + spaCy en_core_web_md; custom recognizers for street addresses (case-sensitive since fix F3), NJ driver licence, date of birth
search_policy	query, k → status + verbatim passages with metadata	Policy Research; the ToT targeted retrieval	Hybrid retrieval with evidence thresholds
propose_redactions	record id, finding ids, chunk ids → per-finding action	Compliance & QA	Accepts chunk ids only, never free-text citations; REDACT requires a supporting chunk
validate_citations	chunk ids (+ excerpts) → exists / verbatim / official / valid	Compliance & QA; ToT critic	Marks non-nj.gov sources invalid (fix F1)

Safety properties:

- every tool is read-only, so none writes records or publishes;
- unknown tools are refused;
- inputs and outputs are validated with Pydantic.

Tested by test_every_mcp_tool_is_read_only, test_unknown_tools_are_refused and test_call_unknown_tool.

Transport evidence:

- python -m app.mcp_client_demo discovers and calls all four tools.
- The Milestone 4 and 5 demos run every agent tool call through the stdio server; their traces record transport mcp.
- tests/test_mcp_integration.py starts the server as a subprocess.
- All of these were re-run successfully in Milestone 7.
- The Milestone 6 evaluation used the in-process executor around the same ToolService to keep runs fast. This is stated in its environment block.

7. ReAct reasoning and tool-use workflow

Status: Implemented as deterministic, rule-based ReAct. LLM-driven ReAct is only partially implemented: the Claude planner was tested with a fake client only.

- **Single agent (Milestone 3, `app/agents/react_agent.py`):** a bounded loop. A planner chooses the next action, an executor (in-process or MCP) runs it, the observation is recorded, and the loop repeats until finish or a limit.
- **Planners:**
  - RulePlanner (used everywhere);
  - ClaudePlanner, which builds tool-use requests for the Claude API but has never been run against the live API, because no API key was configured (tests/test_claude_planner.py uses a fake client).

- **Guardrails:** step limit, repeated-failure stop, and no REDACT without a policy reference (enforced by the schema).
- **In the multi-agent workflow,** each worker runs its own bounded decide → act → observe loop:
  - **Record Analysis:** calls `detect_pii`; on failure, retries (≤ 2 attempts) and never invents findings.
  - **Policy Research:** queries per category. If no passage names the category, it reformulates (≤ 2 queries). For categories OPRA lists outright, it continues until operative statutory text is found (fix F4).
  - **Compliance & QA:** runs `propose_redactions`, then independently validates every citation, classifies evidence (CLEAR / AMBIGUOUS / CONFLICTING / MISSING), and sweeps the text for possibly missed values.

Recorded trace, Scenario A (`docs/milestone_4/execution_traces/scenario_A.md`). Record: *"…My personal phone number is 609-555-0142."*

#	Agent	Tool	Result
1	record_analysis	`detect_pii`	PHONE_NUMBER (0.75)
2	policy_research	`search_policy`	SUFFICIENT: grc_exemptions:0026, opra_statute:0053, opra_statute:0052, grc_handbook:0031
3	compliance_qa	`propose_redactions`	REDACT
4	compliance_qa	`validate_citations`	5/5 valid

Route: `standard` (ToT not invoked). Result: REDACT citing N.J.S.A. 47:1A-5(a), paused for human approval.

8. Semantic retrieval and policy knowledge base

Status: Implemented.

Element	Implementation
Sources	5 official nj.gov documents (`data/policy_documents/manifest.json`): OPRA statute, GRC readable version (Sept 2024, P.L. 2024 c.16); GRC OPRA exemptions list; GRC "Redacting Government Records" page; GRC Handbook for Records Custodians, 8th ed. (Jan 2025); AG Directive 2022-1 (body-worn cameras)
Ingestion	PDF/HTML extraction and cleaning; section-aware chunking (496 chunks), each carrying title, section label (for example `N.J.S.A. 47:1A-5(a)`), page, effective date and source URL. Checkpoint 3.1 ingestion checks the downloaded PDF against the indexed copy by SHA-256
Keyword	BM25
Semantic	ChromaDB with all-MiniLM-L6-v2 (ONNX) embeddings
Hybrid	Reciprocal-rank fusion, k = 4
Evidence gate	Keyword: BM25 ≥ 6.0 plus term-coverage rules. Semantic: cosine ≥ 0.55. Otherwise **INSUFFICIENT**, and the only possible outcome is HUMAN_REVIEW_REQUIRED
Provenance guard	Only passages whose source URL starts with an official nj.gov prefix can support a recommendation. Others are recorded in `excluded_sources` (fix F1)
Untrusted content	Passages are data. Nothing in a passage is executed or followed

Executed example (Checkpoint 3.1, re-run in Milestone 7).

- Query: *"What New Jersey law applies when a public record contains someone's personal telephone number?"*
- Result: rank 2, chunk `opra_statute:0052`, N.J.S.A. 47:1A-5(a), p. 15, cosine 0.6537.
- Excerpt: *"…the custodian thereof shall redact from that record any information which discloses the social security number, credit card number, personal telephone number, or driver license number of any person…"*

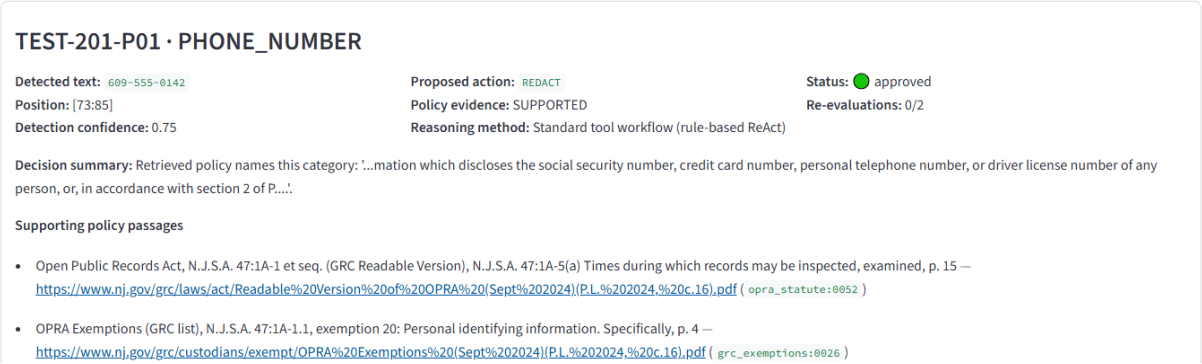


Figure 2. Real reviewer UI (Scenario A): the recommendation shows the source title, statutory section, page, URL and chunk id for each supporting passage.

Missing evidence (Scenario C, IP address).

- Three searches returned passages, but none names IP addresses: two queries plus one targeted retry.
- The result is HUMAN_REVIEW_REQUIRED with **no citation**. The system never writes a citation of its own.

9. Multi-agent orchestration and communication

Status: Implemented.

Agent	Responsibility	Bounded loop
Orchestrator	Plans; routes; merges findings with evidence; decides the route; invokes ToT; handles retries, step limit, escalation, termination	Step limit 30; LangGraph recursion limit
Record Analysis	Detects candidate personal data with offsets and confidence	≤ 2 attempts; ≤ 1 recovery cycle
Policy Research	Retrieves verbatim passages per category; reformulates; performs targeted retrievals on request	≤ 2 queries per category; 1 targeted retry
Compliance & QA	Proposes redactions; validates citations; classifies evidence; sweeps for missed values; acts as the ToT critic	One proposal and one validation pass per round

Coordination:

- **Sequential** where outputs feed inputs: detect → research → merge → QA.
- **Parallel** inside policy research, one search task per category.
- **Conditional** at `evidence_decision`, in priority order: step limit, then detection failure, then MISSING (retry once), then AMBIGUOUS or CONFLICTING (ToT), then standard.

Communication:

- **Through shared typed state only.** Agents never message each other.
- **Appended fields.** List fields (`tool_calls`, `audit`, `transitions`, `escalations`) use `operator.add` reducers and are append-only.
- **Owned fields.** Every other field is written by its owning node.
- **The ToT keeps its own state.** Each thought node has a branch-local copy of evidence, so one branch cannot contaminate another. This is checked by a test and by a safety check over all evaluation runs.

Recorded evidence: five Milestone 4 scenarios, all checks passing, re-run in Milestone 7:

Scenario	Route	Result
A	standard	REDACT, ToT not invoked
B	`tot`	REDACT, score 100
C	targeted retry	escalated, no citations

Scenario	Route	Result
D	ToT	unsupported branch rejected; escalated
D-budget	ToT	tool budget enforced; escalated

10. Selective ToT design and implementation

Status: Implemented (LangGraph subgraph `app/workflows/tot.py`), with the changes listed in the table at the end of this section.

10.1 Why ToT only for ambiguous scenarios

Most findings have one clear answer. For a personal phone number, 47:1A-5(a) requires redaction, and exploring alternatives there only adds cost. Selective ToT keeps the standard path for CLEAR evidence and spends extra reasoning only where an interpretation depends on conditions. For example, a home address is protected only for certain classes of person. The measured cost on ToT cases is 1–3 extra tool calls each.

10.2 The ambiguity trigger

`app/workflows/evidence.py` classifies each finding from retrieved text and the record only:

Class	Rule	Route
CLEAR	Category listed without qualification (phone, driver licence, SSN, month and day of birth), operative language in a retrieved passage, detection confidence ≥ 0.6, no applicable exclusion	Standard path
AMBIGUOUS	Evidence exists but depends on conditions (who the person is, a definition or balancing test, low confidence)	**ToT**
CONFLICTING	A redaction provision and a possibly applicable exclusion (for example "shall not include … telephone number of a public agency")	**ToT**
MISSING	No retrieved passage names the category	One targeted retrieval, then escalate

Measured: ToT activated on 7 of 7 ambiguous scenarios and on 0 of 14 straightforward ones.

10.3 Thought, node and branch representation

- **Thought:** an evidence-based interpretation. Example: "Redact: 47:1A-1.1 protects this home address if the person is a covered person."
- **Node (`ThoughtNode`):**
  - proposed action (REDACT or HUMAN_REVIEW_REQUIRED);
  - provision type;
  - cited chunk ids, with a branch-local copy of their text;
  - assumptions, each FACTUAL or LEGAL with a status (SUPPORTED / UNVERIFIED / REJECTED);
  - unresolved questions;
  - status and score.
- **Branch:** a path from a depth-1 interpretation through verification. The id records it, for example `d1.1>2`.

10.4 Branching factor 3, depth 3, beam width 2

Depth	Stage	What happens

Depth	Stage	What happens
1	INTERPRETATION	Up to 3 candidates built only from retrieved provisions (restricted class, general operative, definition or balancing), always including a conservative HUMAN_REVIEW candidate
2	VERIFICATION	FACTUAL assumptions are checked against the record near the finding. If operative text is missing or a LEGAL assumption is open, the branch may spend the run's **single** targeted retrieval
3	CONCLUSION	Remaining gaps are recorded as unresolved questions

After each depth, at most **2** eligible branches are kept (beam). `ToTConfig` defaults:

- `max_branching=3`, `max_depth=3`, `beam_width=2`, `min_score=70`;
- `max_tool_calls=6`, `time_budget_s=60`.

A test confirms that these limits cannot be raised beyond the design values (`test_tot_limits_cannot_be_raised_beyond_design`).

10.5 Scoring rubric, mandatory checks and critic evaluation

The Compliance & QA Agent acts as the critic. It is **deterministic**, and it validates citations through the `validate_citations` MCP tool.

Criterion	Points	How it is scored
Policy relevance	30	15 if a cited passage names the category. For REDACT: +10 for operative language, +5 for a statutory (primary) source
Citation validity	30	30 if all citations validate. 15 for a REDACT candidate whose valid citations lack operative language. 0 if they could not be validated
Consistency with facts and conditions	25	25 minus 12 for each assumption still UNVERIFIED
Explanation and uncertainty	15	Stated interpretation, unresolved questions listed, human-approval condition attached

Mandatory checks reject a branch immediately:

- a fabricated citation (the chunk does not exist), an altered excerpt, or a non-official source;
- a FACTUAL assumption the record does not support;
- a REDACT branch with no evidence.

Selection constraint. A REDACT branch is selectable only if:

- every assumption is SUPPORTED;
- it cites an operative provision;
- it scores ≥ 70.

Low-confidence detections (< 0.6) add a FACTUAL "detection is correct" assumption that only a reviewer can confirm (fix F2).

10.6 Beam-search selection, pruning and termination

Rule	Behaviour
Pruning	Rejected branches are pruned with the reason recorded; branches beyond the beam get a "beam width" reason
Uncertain branches	Kept as UNVERIFIED, not discarded. This is the Checkpoint 4.1 mitigation for premature pruning
Selection	The highest-scoring selectable branch
Ties	Stronger primary source wins: statute, then GRC list, then handbook. A remaining tie

Rule	Behaviour
	between different actions is an unresolved material conflict and escalates
Termination	Search ends when no open branches remain, depth 3 is reached, or the tool or time budget runs out. When a budget runs out, the event is recorded and nothing unverifiable can be selected

10.7 Missing evidence, unresolved conflicts and human escalation

- If no branch qualifies, the outcome is **ESCALATED**, with the unresolved questions listed.
- A selected REDACT still goes to the human reviewer as a *recommendation*. No branch can authorize disclosure.
- In review, the reviewer can answer FACTUAL conditions (`covered_person`, `not_agency_number`, `detection_correct`) and request re-evaluation. ToT then reruns with those answers.
- A reviewer cannot satisfy a LEGAL condition or create a citation (`test_reviewer_facts_are_factual_input_only`).

10.8 Real execution: Scenario B (retired judge's home address)

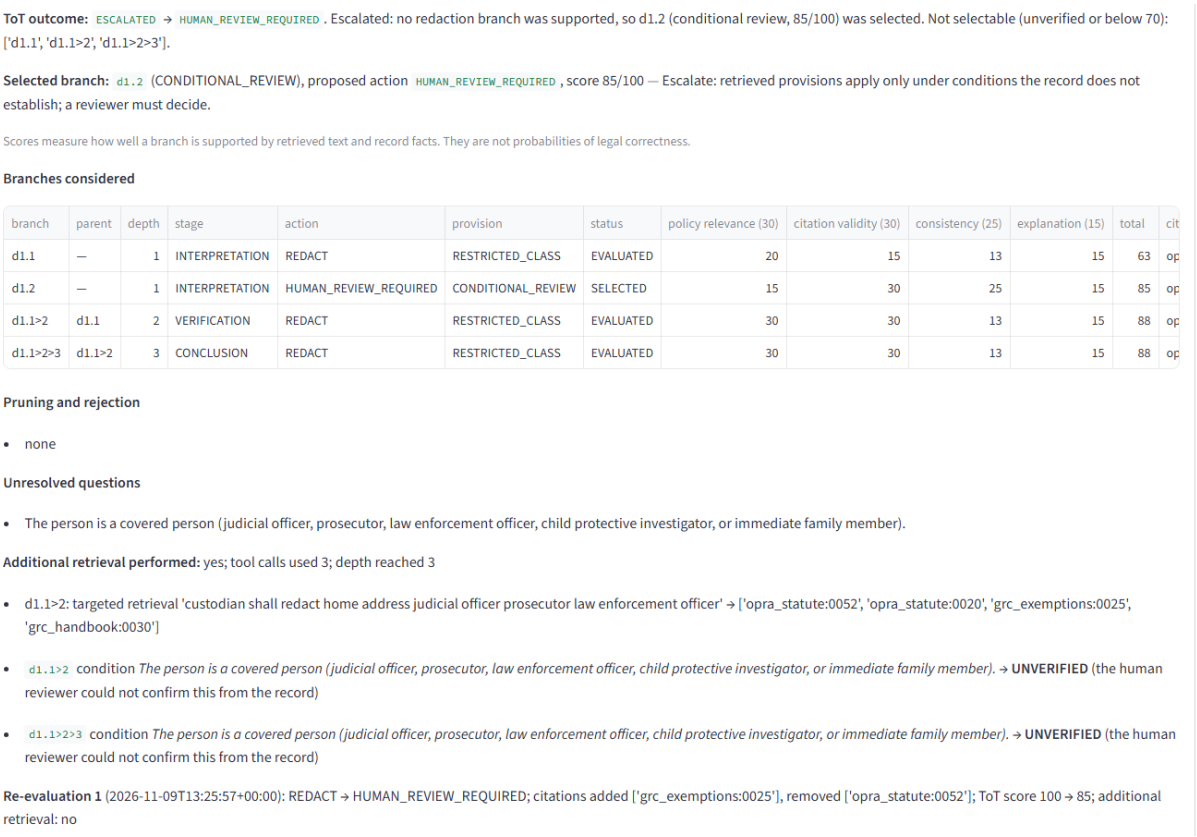
Source: `docs/milestone_4/execution_traces/scenario_B.md`.

Branch	Depth	Action	Provision	Rel	Cit	Cons	Expl	Total	Status
d1.1	1	REDACT	RESTRICTED_CLASS	20	15	13	15	63	evaluated (covered status unverified)
d1.2	1	HUMAN_REVIEW_REQUIRED	CONDITIONAL_REVIEW	15	30	25	15	85	evaluated
d1.1>2	2	REDACT	RESTRICTED_CLASS	30	30	25	15	**100**	**selected**

How the selection happened:

- At depth 2, the covered-person assumption was SUPPORTED by the record ("retired Superior Court judge").
- The one targeted retrieval (*"custodian shall redact home address judicial officer prosecutor law enforcement officer"*) added the operative 47:1A-5(a) passage `opra_statute:0052`.

Reviewer feedback in Milestone 5. The reviewer answered `covered_person = unconfirmed`. ToT reran, both REDACT branches became non-selectable (88, unverified), d1.2 (85) was selected, and the item escalated.



ToT outcome: `ESCALATED` → `HUMAN_REVIEW_REQUIRED` . Escalated: no redaction branch was supported, so d1.2 (conditional review, 85/100) was selected. Not selectable (unverified or below 70): ['d1.1', 'd1.1>2', 'd1.1>2>3'].

Selected branch: `d1.2` (CONDITIONAL_REVIEW), proposed action `HUMAN_REVIEW_REQUIRED` , score 85/100 — Escalate: retrieved provisions apply only under conditions the record does not establish; a reviewer must decide.

Scores measure how well a branch is supported by retrieved text and record facts. They are not probabilities of legal correctness.

Branches considered

branch	parent	depth	stage	action	provision	status	policy relevance (30)	citation validity (30)	consistency (25)	explanation (15)	total	cit
d1.1	—	1	INTERPRETATION	REDACT	RESTRICTED_CLASS	EVALUATED	20	15	13	15	63	op
d1.2	—	1	INTERPRETATION	HUMAN_REVIEW_REQUIRED	CONDITIONAL_REVIEW	SELECTED	15	30	25	15	85	op
d1.1>2	d1.1	2	VERIFICATION	REDACT	RESTRICTED_CLASS	EVALUATED	30	30	13	15	88	op
d1.1>2>3	d1.1>2	3	CONCLUSION	REDACT	RESTRICTED_CLASS	EVALUATED	30	30	13	15	88	op

Pruning and rejection

- none

Unresolved questions

- The person is a covered person (judicial officer, prosecutor, law enforcement officer, child protective investigator, or immediate family member).

Additional retrieval performed: yes; tool calls used 3; depth reached 3

- d1.1>2: targeted retrieval 'custodian shall redact home address judicial officer prosecutor law enforcement officer' → ['opra_statute:0052', 'opra_statute:0020', 'grc_exemptions:0025', 'grc_handbook:0030']
- `d1.1>2` condition *The person is a covered person (judicial officer, prosecutor, law enforcement officer, child protective investigator, or immediate family member).* → **UNVERIFIED** (the human reviewer could not confirm this from the record)
- `d1.1>2>3` condition *The person is a covered person (judicial officer, prosecutor, law enforcement officer, child protective investigator, or immediate family member).* → **UNVERIFIED** (the human reviewer could not confirm this from the record)

Re-evaluation 1 (2026-11-09T13:25:57+00:00): REDACT → HUMAN_REVIEW_REQUIRED; citations added ['grc_exemptions:0025'], removed ['opra_statute:0052']; ToT score 100 → 85; additional retrieval: no

Figure 3. Real reviewer UI (Scenario B after re-evaluation): ToT outcome, selected branch, every branch with its actual scores, unresolved questions and the targeted retrieval. Scores carry the note that they are not probabilities of legal correctness.

10.9 Implemented vs proposed (Checkpoint 4.1)

4.1 design element	Status
ToT only when evidence is ambiguous or conflicting	**Implemented**
Thought / node / branch representation (evidence, assumptions, action, open questions)	**Implemented**
Branching 3, depth 3, beam 2, beam search	**Implemented** and enforced
30/30/25/15 rubric; minimum 70	**Implemented**
Reject fabricated citations and unsupported factual assumptions	**Implemented** (2 rejections in the evaluation)
Ties go to stronger primary-source evidence; unresolved conflicts go to human review	**Implemented**. Covered by unit tests only, because no scenario produced a tie
Uncertain branches marked, not pruned; one targeted retrieval before pruning	**Implemented** (13 uncertain branches preserved)
Time, depth and tool-call limits	**Implemented**
Critic **agent** (CrewAI + LangChain) assessing relevance and explanation quality	**Changed:** a deterministic rubric inside the Compliance & QA Agent. Explanation quality is checked structurally only
Thought generator in CrewAI; controller in LangChain	**Changed:** a single LangGraph subgraph (generate → evaluate → prune → select)
Example branch types A (redact) / B (exception) / C (escalate)	**Implemented in spirit.** Exceptions such as "use by any government agency" are surfaced as open questions, not reasoned about as separate branches

11. Human feedback, approval, redaction and auditability

Status: Implemented.

The workflow pauses at `human_review` (LangGraph `interrupt()`). A `ReviewSession` snapshot drives the Streamlit app (`streamlit run app/review_app.py`).

Capability	Behaviour
Per-finding decisions	Approve (refused for escalated items: nothing supported to approve); reject with an optional reason; modify exact spans (spans must be confirmed)
Re-evaluation	Reviewer-requested only, never automatic on rejection. Takes a structured reason, optional factual answers and an optional extra query. Runs through the same Policy Research and QA agents, plus ToT if still ambiguous. Limited to **2 per finding**, then MANUAL_RESOLUTION_REQUIRED. Records a before/after comparison
Reviewer-added redaction	For text the detector missed. QA flags possible misses (spelled-out or 7-digit numbers) as escalations. The reviewer selects the exact text, a category and a reason, then confirms. It is recorded as a human decision with no system policy basis claimed
Finalization	A separate step. Blocked, with reasons recorded, unless every finding is resolved, every escalation is acknowledged, a named reviewer authorizes, the event chain verifies, the source file is unchanged, and span validation passes
Deterministic redaction	The derivative is built by slicing the original text at validated offsets. Overlapping spans merge. Untouched segments are verified byte-identical. A redacted value repeated elsewhere blocks finalization
Originals	Never modified. The derivative goes to `outputs/derivatives/…`. The original record's hash is checked
Audit	Append-only, hash-chained review events (actor, UTC time, payload, previous hash). The audit report contains the workflow snapshot, tool calls, ToT tables, re-evaluations, decisions and finalization. `published: false`



Figure 4. Real reviewer UI (Scenario C): finalization refused for an escalated finding with no policy basis.

Full-page captures of all three scenarios are in `docs/final_capstone/screenshots/`. Recorded sessions and audit reports are in `docs/milestone_5/`.

12. Evaluation methodology

Details: evaluation_summary.md.

Item	Choice
Dataset	22 synthetic scenarios, 19 expected findings: clear categories, detector edge cases, decoys, missing evidence, ambiguous and conflicting policy, tool faults, prompt injection (in a retrieved document and in the record), reviewer rejection
Configurations	A: `enable_tot=False`. B: `enable_tot=True`. Everything else identical
Runs	44 (one trial each; agents are deterministic)
Metrics	Detection precision and recall; recommendation in acceptable set and preferred; grounded citation accuracy; unsupported redactions; workflow completion; end-to-end success; escalation recall and false escalation rate; tool-failure safe handling and recovery; latency; tool calls; ToT activation, branch validity, pruning, termination, invariants; 16 safety checks
Reviewer	Scripted from scenario definitions
Integrity	Expected outcomes were not changed to make results pass. The pre-fix run is preserved. All metrics are recomputable from raw traces (`python -m app.evaluation.verify`, 33/33 match)

13. Actual evaluation results

Metric	A: ReAct-only	B: ReAct + ToT
Detection precision	1.000 (17/17)	1.000 (17/17)
Detection recall	0.895 (17/19)	0.895 (17/19)
Recommendation in acceptable set	1.000 (17/17)	1.000 (17/17)
Preferred recommendation	0.882 (15/17)	0.941 (16/17)
Grounded citation accuracy	1.000 (6/6)	1.000 (7/7)
Unsupported redactions	0	0
Workflow completion	22/22	22/22
End-to-end success	22/22	22/22
Escalation recall / false escalation rate	11/11 / 0/9	11/11 / 0/9
Tool failure: safe handling / full recovery	3/3 / 3/3	3/3 / 3/3
Latency mean / median / p95 (s)	0.377 / 0.371 / 1.059	0.388 / 0.364 / 0.987
Mean tool calls per scenario	3.59	4.18
LLM calls / cost	0 / $0	0 / $0

ToT (B):

- **Activation:** 7 TP, 0 FP, 0 FN, 14 TN (S16 not scored).
- **Branches:** 8 runs; 25 branches (20 supported by official evidence); 2 rejected for unsupported factual assumptions; 0 for fabricated citations; 13 uncertain branches preserved, 0 lost.
- **Limits:** 0 invariant violations.
- **Outcomes:** 1 fully supported REDACT, 7 escalations.
- **Cost:** 13 extra tool calls.

Detection misses. S04 (phone spelled out in words) and S05 (7-digit number). Both still succeed end to end, because QA flags them and the scripted reviewer adds the redaction. The detector itself does not catch them.

Defects found and fixed (pre-fix B: 2 unsupported redactions, 3 false activations):

#	Defect	Fix
F1	Injected non-official text could be cited	Provenance guard
F2	ToT ignored low detection confidence	Reviewer-only "detection is correct" condition
F3	Case-insensitive address regex	Recognizer made case-sensitive
F4	Retrieval missed operative text for a listed category	Research continues until operative text is found

14. ReAct-only vs ReAct+ToT comparison

Details: react_vs_tot_comparison.md.

Question	Measured answer
Decision quality	+1 preferred decision (S12). All 17 matched findings were acceptable in both configurations
Evidence coverage	S12 gained the operative 47:1A-5(a) citation through the targeted retrieval. Grounded accuracy was 1.000 in both
Escalation behaviour	No change (11/11, 0/9)
Safety	No difference after the fixes. Worse with ToT before the fixes (2 unsupported redactions)
Cost	+13 tool calls in total (1–3 per ToT case); +0.111 s mean per ToT case; $0
Conclusion	**Inconclusive for general accuracy.** Selective ToT did no harm after the fixes and helped one ambiguous case. A rubric score measured consistency, not correctness: before the fixes, ToT rated two wrong selections "fully supported"

15. Safety controls and failure handling

Risk	Control	Evidence
Fabricated citations	Chunk-id-only citations; `validate_citations`; QA veto; ToT rejection	5 guardrail tests; safety check "every citation is an existing chunk of an official source" (44/44 runs)
Unsupported legal conclusions	INSUFFICIENT or MISSING → HUMAN_REVIEW_REQUIRED; schema forbids REDACT without a reference; REDACT selection needs verified assumptions	4 guardrail tests; Scenario C
Prompt injection	Rule-based agents; provenance guard; record text never treated as instructions	S20: injected document never cited and injected citation `47:1A-99` never repeated (A and B). S22: instructions inside the record ignored; expected outcome in both configurations
Modifying originals	Read-only tools; hash checks; derivative path checks	3 tests; safety check over 44 runs
Finalization without approval	`interrupt()`; named reviewer plus explicit authorization; blockers	7 tests; Scenario C blocked
Unbounded loops	Step limit, retry limits, ToT depth, beam and tool budgets, re-evaluation limit	7 tests; 0 invariant violations
Tool failure	Timeouts (`ToolCaller`); retry, then escalate; failed policy search becomes MISSING, never invented policy	S17–S19: 3/3 safe handling and recovery
Cross-branch contamination	Branch-local evidence copies	Test plus safety check
Secrets and sensitive data	`.env` git-ignored; no keys needed; synthetic-only loader; reserved example values (555-01xx phones, RFC 5737 IPs, example.com)	Milestone 7 scan: final_verification.md

16. Limitations, lessons learned and future improvements

Full list: limitations.md.

Limitations

- **No LLM in the loop.** The agents are deterministic, so the results do not speak to LLM planning, LLM hallucination or LLM injection resistance.

- **Detector gaps.** Recall is 17/19. Spelled-out and 7-digit numbers are flagged for the reviewer, not detected; other formats would be missed.
- **Small evaluation set.** 22 self-authored scenarios, 7 ToT cases, and a simulated reviewer.
- **Narrow corpus.** Five documents; retrieval is sensitive to wording.
- **Prototype operation.** Text only; single local reviewer; no access control; in-memory graph checkpoints.

Lessons learned

1. **A baseline comparison finds what unit tests do not.** Running A against B exposed four defects. Two of them produced harm only when ToT was on.
2. **Scores are not correctness.** ToT rated two wrong selections as fully supported. The protection came from mandatory evidence checks and the human gate, not from the score.
3. **Controlling evidence beats generating more reasoning.** The provenance guard and operative-text requirement did more for safety than additional branches.
4. **Deterministic first.** Rule-based agents made every decision reproducible and every metric recomputable. That made the evaluation trustworthy, at the cost of LLM flexibility.
5. **Escalation is a feature.** Most ambiguous cases end correctly with "human review required"; the system's job there is to package the evidence and questions.

Future improvements (not implemented)

- An LLM thought generator or critic behind the same validation gates, evaluated against these scenarios plus independently written ones.
- Detector improvements for spelled-out and partial numbers.
- More policy sources (agency rules, GRC decisions) with provenance checks.
- Synthetic audio and video (transcription, frame redaction): the original Stage 8.
- Multi-user review with authentication and second-reviewer sign-off.
- A usability study with records staff.

Appendix A. Feature verification matrix

Verified in Milestone 7 against code, tests and recorded runs. "Run in M7" means it was re-executed during Milestone 7 (see final_verification.md).

Feature	Status	Evidence	Run in M7
Synthetic data + loader refusing non-synthetic files	Implemented	`app/data_loader.py`; `tests/test_data_loader.py` (17)	Yes (`python -m app.main`)
Pydantic schemas (findings, references, recommendations, decisions, state)	Implemented	`app/schemas.py`, `app/workflows/state.py`; `tests/test_schemas.py` (26)	Yes (suite)
Policy ingestion from nj.gov with provenance	Implemented	`app/policies/ingest.py`; `docs/checkpoint_3_1/01_*`	**Network download failed in M7** (no DNS in this environment); local-copy path passed
Keyword (BM25), semantic (ChromaDB) and hybrid retrieval with thresholds	Implemented	`app/policies/`; `tests/test_policy_*`	Yes
PII detection (Presidio + custom recognizers)	Implemented; recall 17/19	`app/tools/pii.py`; `tests/test_tools_pii.py`	Yes
MCP server with 4 read-only tools (official SDK, stdio)	Implemented	`app/mcp_server.py`; `tests/test_mcp_integration.py`; `mcp_client_demo`	Yes
Single ReAct agent, rule-based planner	Implemented	`app/agents/`; `tests/test_agent.py` (13)	Yes (`agent_demo`, local and MCP)
LLM-driven ReAct (Claude planner)	**Partially implemented:** fake client only, never run live	`tests/test_claude_planner.py`	Not run (no API key)
Four-agent LangGraph orchestration	Implemented	`app/workflows/`; `tests/test_m4_workflow.py`, `test_m4_scenarios.py`	Yes (`m4_demo`, 5/5 scenarios)
Standard cases bypass ToT	Implemented	Scenario A; 14/14 TN in the evaluation	Yes
Ambiguous cases invoke ToT	Implemented	Scenario B; 7/7 TP	Yes
ToT branching, depth, beam, scoring, pruning, budgets	Implemented	`tests/test_m4_tot.py` (18); 0 invariant violations	Yes
ToT tie-breaking and beam pruning in real scenarios	Implemented; **unit-test evidence only**	`test_beam_width_limits_retained_candidates`, `test_tie_goes_to_stronger_primary_source`, `test_unresolved_material_conflict_escalates`	Yes (suite)
LLM critic agent for ToT	**Not implemented** (changed to a deterministic critic)	—	—
Missing evidence → safe escalation	Implemented	Scenario C; S10, S11, S18	Yes
Human review interrupt / resume	Implemented	`test_only_human_approval_authorizes_disclosure`	Yes
Streamlit reviewer UI	Implemented	`app/review_app.py`; `tests/test_m5_app.py` (6); screenshots	Yes (screenshots captured in M7)
Re-evaluation with structured feedback (≤ 2)	Implemented	`tests/test_m5_review.py`; Scenario B	Yes (`m5_demo`)
Reviewer-added redaction for missed text	Implemented	`tests/test_m6_manual_redaction.py` (11 test functions)	Yes (suite)
Deterministic redaction + validation	Implemented	`app/review/redact.py`; `tests/test_m5_redact.py` (9)	Yes
Finalization blockers / explicit authorization	Implemented	`tests/test_m5_review.py`; Scenario C	Yes
Hash-chained audit log and audit report	Implemented	`app/review/session.py`, `report.py`	Yes
Prompt-injection resistance (retrieved document, record	Implemented (rule-based agents)	S20, S22; F1 tests	Yes (`evaluation.run`)

Feature	Status	Evidence	Run in M7
text)			
Evaluation harness, metrics, A/B comparison	Implemented	`app/evaluation/`; `tests/test_m6_evaluation.py` (17)	Yes (rerun identical; verify 33/33)
Audio / video redaction	**Deferred**	—	—
Multi-user access control, persistent checkpoints	**Not implemented**	—	—

Appendix B. Differences from the original proposal

Area	Originally planned (proposal and checkpoints)	Implemented	Why changed	Effect on reliability and scope
Record types	Video, audio, transcripts, documents	Text records only	Time; text covers the core reasoning; synthetic media pipeline not needed to demonstrate agents	Narrower scope; transcription tool and OpenCV/FFmpeg deferred (Stage 8)
Reasoning engine	ReAct with an LLM (implied by 2.1/3.1/4.1)	Deterministic rule-based ReAct; Claude planner built but untested live	No API credentials; reproducible, testable behaviour; avoid hallucination risk in a compliance task	Every result reproducible and recomputable; no evidence about LLM behaviour
Frameworks for ToT	CrewAI + LangChain + MCP	LangGraph + MCP	One orchestration framework (LangGraph was already the workflow engine); fewer dependencies	Simpler state handling; branch state isolated by design
ToT critic	"Critic agent" assessing relevance and explanation quality	Deterministic rubric inside Compliance & QA	Without an LLM, explanation quality cannot be judged semantically	Explanation criterion is structural; mandatory checks unchanged
ToT adoption	4.1 drafts: first "no ToT", then "selective ToT"	Selective ToT	Ambiguous provisions (restricted classes, exclusions) needed structured alternatives	One measured gain (S12); pre-fix harms found and fixed
Agent coordination	Record analysis and policy research coordinated, partly in parallel (5.1)	Research runs after detection; parallel across categories	What to research depends on what was found	Clearer data flow; no wasted retrieval
Feedback on rejection	"If a proposed redaction is rejected, the agent can review the evidence and policy again" (proposal)	Rejection is final; re-evaluation is an explicit reviewer request, ≤ 2 per finding	Avoid automatic loops; keep the human in control	Bounded, auditable feedback
Missed-PII check	"A final review step can also check the proposed output for possible missed sensitive information"	QA sweep for two missed formats + reviewer-added redaction	Detector gaps found in evaluation (S04, S05)	Gaps flagged, not solved
Evaluation transport	Not specified	In-process executor for the evaluation; MCP for demos and integration tests	Speed and determinism of 44 runs	Same `ToolService`; MCP covered separately
Additions not in the proposal	—	Provenance guard; low-confidence condition; hash-chained events; finalization blockers; metric recomputation script	Found necessary during evaluation and review design	Stronger safety guarantees

Appendix C. Reproducibility

- **Final evaluation commit:** `7a1a875` ("Milestone 6: evaluation, guardrails, ToT validation, reviewer-added redactions").

- **Environment:** Python 3.14.7 on Windows 10. Package versions are pinned in `requirements.txt`; key ones are langgraph 1.2.20, chromadb 1.5.9, presidio-analyzer 2.2.364, mcp 2.3.0, streamlit 1.66.0.
- **Clean-clone check (Milestone 7):** install exit 0 in 272 s, then 314 tests passed.

```
python -m venv .venv && .venv/Scripts/activate        # Windows; use source .venv/bin/activate
elsewhere
pip install -r requirements.txt
python -m pytest                                       # 314 passed
python -m app.checkpoint_3_1                           # real-source retrieval demo (needs network; --
offline uses the local copy)
python -m app.m4_demo                                  # four-agent workflow + ToT scenarios over MCP
python -m app.m5_demo                                  # review scenarios A/B/C
streamlit run app/review_app.py                        # reviewer UI
python -m app.evaluation.run --tag rerun               # evaluation (does not overwrite the committed
final results)
python -m app.evaluation.verify                        # recompute committed metrics from raw traces
```

Appendix D. Evidence index

Evidence	Location
Real-source retrieval demo	`docs/checkpoint_3_1/`
Workflow and ToT traces	`docs/milestone_4/execution_traces/`
Review sessions, audit reports	`docs/milestone_5/review_scenarios.md`, `docs/milestone_5/scenarios/`
Evaluation results, ToT comparison, safety tests, raw traces	`docs/milestone_6/`
Screenshots of the running app	`docs/final_capstone/screenshots/`
Milestone 7 verification logs	`docs/final_capstone/verification_logs/`
Architecture sources	`docs/final_capstone/architecture/`